CS407 Group Project 4: Subscription Churn & User Engagement - Streaming Platform

Business Context

Subscription-based streaming platforms depend on recurring revenue and sustained engagement. When activity declines, subscriptions are paused, or customers encounter billing and service problems, users may be more likely to cancel. In this project, your team will build a classification model to predict subscription churn from customer, payment, and engagement behavior. The model can support targeted retention strategies such as content discovery, service recovery, or plan adjustments. This project provides experience with behavioral feature engineering and business-driven ML.

Dataset Link

https://www.kaggle.com/competitions/streaming-subscription-churn-model/data

Project Objectives

1. Handle missing values, parse date fields, encode categorical variables, and perform exploratory data analysis (EDA).
2. Engineer meaningful features from listening behavior, playlists, social activity, subscription history, payments, notifications, and support interactions.
3. Train multiple classification models, starting with a baseline such as logistic regression.
4. Evaluate precision, recall, F1-score, PR-AUC, and ROC-AUC; compare decision thresholds and class-imbalance strategies.
5. Recommend evidence-based retention actions for different types of at-risk subscribers.

Project Flow and Deliverables

1. Data Preprocessing - Deliverable 1: Clean, leakage-safe dataset with EDA insights and visualizations.
2. Feature Engineering - Deliverable 2: Dataset with interpretable engagement, tenure, and interaction features.
3. Model Selection and Training - Deliverable 3: Baseline and advanced classifiers with reproducible pipelines.
4. Evaluation and Interpretation - Deliverable 4: Cross-validated metrics, threshold analysis, imbalance comparison, and explanations.
5. Final Deliverables - Deliverable 5: Notebook, report, and presentation connecting findings to targeted retention strategies.

Hint

Useful features may include estimated sessions per week, unique-song ratio, skip behavior, playlist and social engagement, subscription pauses, support-contact frequency, notification response, and customer tenure. Exclude identifiers and information recorded after churn, and fit preprocessing or resampling only within training folds. Treat this as an instructional dataset rather than evidence about Netflix, Spotify, or any other named company.